SOMIN

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LECTURE 3 · PART 1 · ITMO UNIVERSITY · 21 OCTOBER 2026

Information Retrieval & Recommender Systems

From the inverted index to generative recommenders.
Part 2, next week: the networks underneath.

Prof. Aleks Farseev · Daniel Onishchenko · Jack Zhao

Lecturer · Tutors



RECAP · YOUR HOMEWORK

You profiled ten authors. Now: what should each one see?

01

FROM THE WORDS

What the posts said

Topics, job, tone, the tools they mention. Evidence you can point to — and so can a classifier.

02

FROM ASSUMPTIONS

What you filled in

Age, seniority, gender — guessed from style, photos or names. A model does the same and calls it a feature.

03

THE NEXT QUESTION

What would you show them?

A job post? A competitor's article? A new connection? Choosing is today's lecture: retrieval and recommendation.

WHY THE QUESTION FLIPS

Lecture 2 built the profile. A profile earns money only when it turns into a ranked list: a search result, a feed, an ad, a “people you may know”. That ranking is the product.

SOURCE Homework from Lecture 2 (ten authors from the Lab 1 data); Farseev, ITMO SMA Lecture 2.

LECTURE 3, PART 1 · OUTLINE

Four questions for today.

01

The matching problem

Search and feeds: one problem, two directions.

02

Classic retrieval

From the inverted index to BM25.

03

Neural search and RAG

Embeddings, rerankers — and LLMs that search.

04

Recommender systems

From collaborative filtering to generative recommenders.

Part 2, next week: networks and graphs. PageRank, weak ties, influence, communities and graph neural networks — the graph underneath search and feeds.

SOURCE ITMO SMA modules 2 (text analysis & retrieval), 7 (graph theory & social networks) and 9 (LBSN & recommendation), updated for 2026.

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01

SECTION 1

The matching problem

META · Q1 2024 EARNINGS CALL

“More than 50% of the content people see on Instagram is now AI recommended.”

And about 30% of Facebook feed posts. The feed stopped being a list of friends' posts. It became a ranking problem — the one search engines solve.





MATCHING · TWO DIRECTIONS

Search pulls. Recommendation pushes. Same engine.

SEARCH · PULL

You say what you want

A query states the need; the system ranks documents by relevance to it. Google, a shop's search box, TikTok search, ChatGPT search.

RECOMMENDATION · PUSH

The system guesses

No query. History, profile and context stand in for one. The For You feed, “people you may know”, “customers also bought”.

THE SHARED CORE

score(u, q, i) → sort

- **u** the user — Lecture 2's profile
- **q** the query, or the context: time, place, device
- **i** the item: a page, post, video, product or person

Search fills q with typed words; a feed fills it with context. After that, the machinery is shared.

THE LINE IS BLURRING IN 2026

- ✓ **Search moved into feeds** TikTok and Instagram are search engines for many users.
- ✓ **Feeds answer unasked queries** AI answers arrive before you finish typing.

SOURCE Farseev, ITMO SMA Module 2 (retrieval vs classification); Leskovec, Stanford CS246 “Recommender Systems I” (Winter 2026).

MATCHING · 2026 IN NUMBERS

Most of what people see was ranked by a model.

5T+

SEARCHES A YEAR

On Google alone (March 2025). Google holds 89.9% of web search worldwide (Sept 2026).

2.5B

AI OVERVIEWS USERS

Every month (Google I/O, May 2026). AI Mode passed 1 billion monthly users within a year.

~80%

OF NETFLIX HOURS

Influenced by recommendations. Netflix valued personalisation at over $1B a year in reduced churn (2015).

70%

OF YOUTUBE WATCH TIME

Driven by recommendations, according to YouTube's product chief (CES, 2018).

READ IT LIKE AN ANALYST

The most cited shares are old: platforms stopped publishing them once feeds became fully recommended. Meta's latest public figure — more than half of Instagram content is AI-recommended — dates from 2024. Treat any single “% from recommendations” as a lower bound with a date.

SOURCE Google via Search Engine Land (Mar 2025); StatCounter (Sep 2026); Google I/O 2026 keynote; Gomez-Uribe & Hunt, ACM TMIS 2015; N. Mohan at CES 2018 (CNET/Tubefilter).

MATCHING · THE UNIVERSAL FUNNEL

Every system is a funnel: retrieve cheaply, rank carefully.



Every page, post, video or ad that could be shown.

Cheap and fast: inverted index, vector search, graph walks. Recall matters.

A heavy model scores each candidate: click, watch, like, buy.

Diversity, freshness, policy, ads — then the screen.

X / TWITTER · 2023 CODE

~500M posts a day → ~1,500 candidates per request → ranked by a ~48M-parameter network.

META ANDROMEDA · 2024

Tens of millions of ads → a few thousand candidates per request, before ranking.

YOUTUBE · 2016

Millions of videos → hundreds of candidates → dozens shown, ranked by expected watch time.

SOURCE Twitter Engineering (31 Mar 2023); Meta Engineering, “Meta Andromeda” (2 Dec 2024); Covington, Adams & Sargin, RecSys 2016.

MATCHING · INPUTS

What a matching system knows: five kinds of signal.

01

Query & context

Words typed; time, place, device; the page you are on.

02

The user

Lecture 2's profile: interests, demographics, a user embedding.

03

The item

Text, image, video, price, freshness, the creator.

04

The graph

Who follows whom, what is bought together, which pages link.

05

Feedback

Clicks, watch time, likes, skips — and negatives: block, mute, report.

FEEDBACK HAS TWO FLAVOURS

- **Explicit** Stars, thumbs, ratings. Rare and biased toward strong opinions.
- **Implicit** Clicks, dwell, completions. Plentiful and noisy.
- **Missing ≠ negative** An item you never saw is not an item you disliked.

SOURCE Leskovec, Stanford CS246 (Winter 2026); xAI, x-algorithm repository (Jan 2026) for negative-action signals.

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02

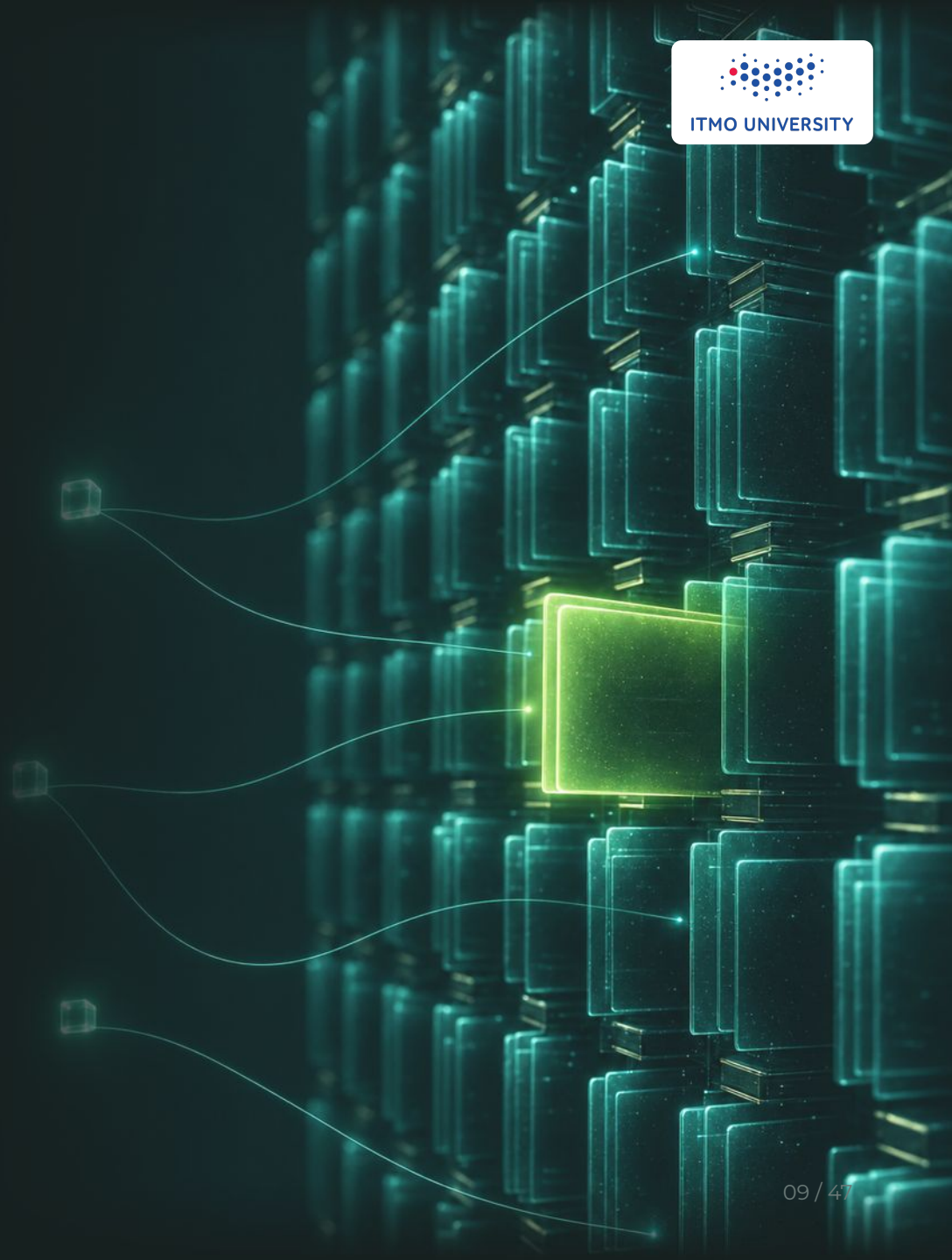
SECTION 2

Information retrieval: the classic engine

GOOGLE · MARCH 2025

“More than 5 trillion searches a year — over an index of hundreds of billions of pages.”

The core of every search engine fits in four ideas: an inverted index, a scoring function, link and behaviour signals, and a way to measure the result.





RETRIEVAL · DEFINITION

Finding the few documents that satisfy a need.

> **“Information retrieval is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections.”**
>
> Manning, Raghavan & Schütze, Introduction to Information Retrieval (2008)

- **Unstructured.** Free text, images, video — not database rows with a schema.
- **Large.** Google's index: hundreds of billions of pages, over 100 million GB.
- **Ranked.** Relevance is a degree, so the output is an ordered list, not a set.

NEED — **In your head**

“A cheap way to fly to Singapore for the November conference.”

↓

QUERY — **What you type**

cheap flights SIN nov — short, vague, 2–3 words.

↓

RESULTS — **What the engine returns**

A ranked list, an AI answer, or both.

↓

RELEVANCE — **Judged against the need**

Not the query. A perfect match for the words can still be useless.

SOURCE Manning, Raghavan & Schütze, IIR ch. 1 (nlp.stanford.edu/IR-book); Google, How Search Works — Organizing information.

RETRIEVAL · RECAP FROM LECTURE 2

You built half a search engine last week.

01	02	03	04	05
Tokenise	Stop list	Stem	tf.idf	Vector

WHAT IS STILL MISSING FOR WEB SCALE

- **Speed.** Comparing a query with every document is impossible at hundreds of billions of pages.
- **Scoring.** Raw counts reward repetition and long documents.
- **Meaning.** “He retrieved his money” matched “information retrieval”.

Today's fixes: the inverted index (speed), BM25 (scoring), embeddings (meaning).

LECTURE 2: SIM(Q, D) = Σ $Q_t \cdot D_t$

D1 3

D2 1

D3 1

Q = “information retrieval”. D3 (“he retrieved his money from the safe”) tied with D2 only because the stemmer turned “retrieved” into “retriev”.

SOURCE Worked example from Farseev, ITMO SMA Module 2 (“A retrieval example”) and Lecture 2, slide 15.

The inverted index: from documents to terms.



DOCUMENTS

D1	information retrieved by VS retrieval methods
D2	information theory forms the basis for probabilistic methods
D3	he retrieved his money from the safe

TERM	DF	POSTINGS (DOC IDS)
info	2	1, 2
retriev	2	1, 3
method	2	1, 2
theory	1	2
money	1	3
safe	1	3

WHY IT WORKS

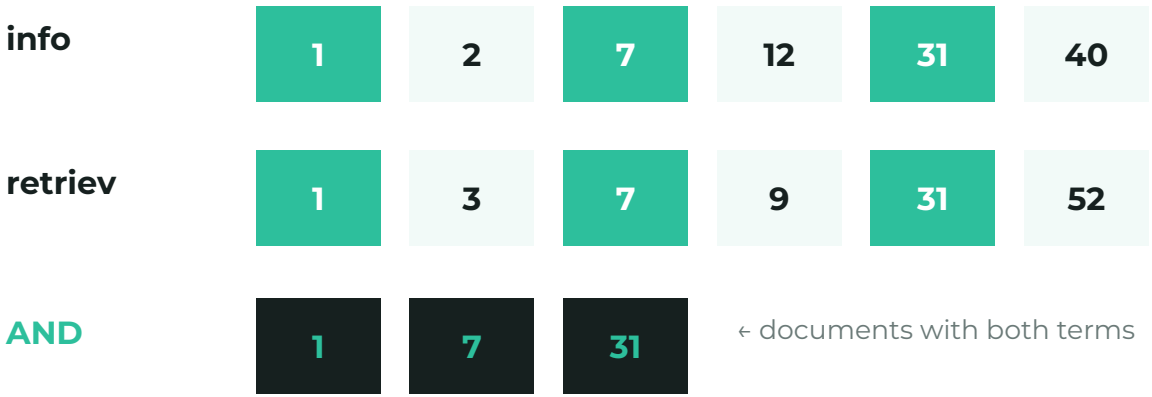
A query only touches the postings of its own terms.

Two query words mean two short lists — not hundreds of billions of documents. Postings are sorted by document ID, compressed, and split across thousands of machines.

SOURCE IIR ch. 1 — "A first take at building an inverted index"; documents from ITMO SMA Module 2.

RETRIEVAL · ANSWERING A QUERY

Answering a query is merging two sorted lists.



- **Two pointers.** Compare the heads; advance the one with the smaller ID; record matches.
- **Cost O(x + y).** Linear in the two list lengths — independent of the collection size.

HOW REAL ENGINES GO FASTER

- **Rarest term first.** Start from the shortest list; every later step can only shrink the result.
- **Skip pointers.** Jump ahead in long lists instead of walking every entry.
- **Safe early stopping.** WAND and MaxScore skip documents that cannot reach the top 10.
- **Sharding.** Each machine holds a slice of the index; a query fans out to all of them.

SOURCE IIR ch. 1–2 (postings intersection, skip pointers); Turtle & Flood 1995 (MaxScore); Broder et al., CIKM 2003 (WAND).

RETRIEVAL · SCORING

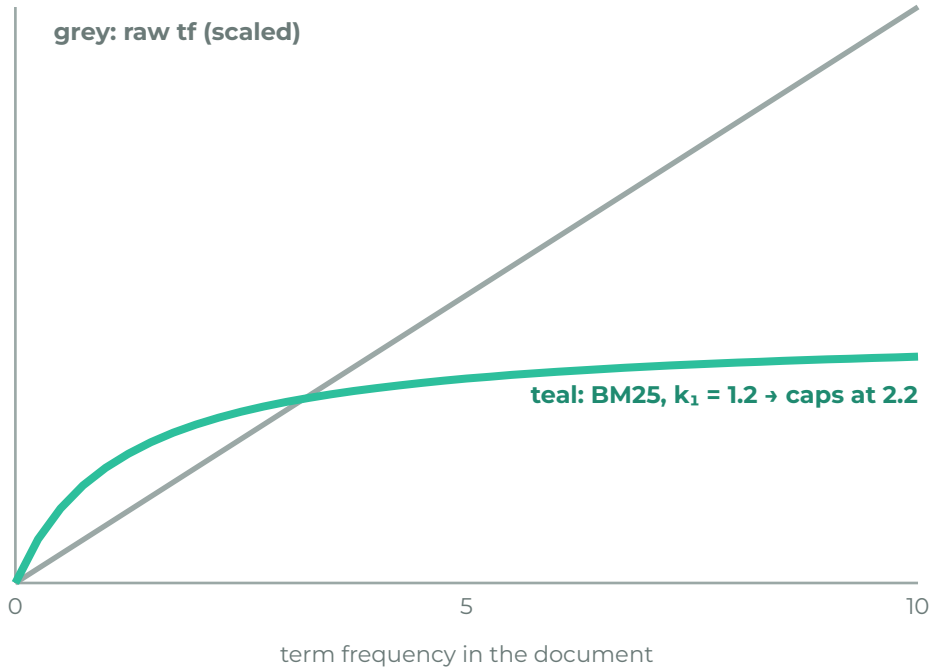
BM25: tf.idf with two corrections.

OKAPI BM25 · ROBERTSON ET AL., TREC-3 (1994)

$$\text{score}(D,Q) = \Sigma\ \text{idf}(t) \cdot \text{tf}\cdot(k_1+1) \,/\, (\text{tf} + k_1\cdot(1 - b + b\cdot|D|/\text{avgdl}))$$

Saturation $k_1 \approx 1.2–2$	The 10th “information” adds little; the 100th adds almost nothing.
Length $b \approx 0.75$	A long document gets no credit just for containing more words.
Rarity idf	Rare terms carry the weight — Spärck Jones's 1972 idea, kept intact.

WEIGHT OF ONE TERM AS ITS COUNT GROWS



SOURCE Robertson et al., “Okapi at TREC-3” (1994); Robertson & Zaragoza, FnTIR 2009; Spärck Jones, J. Documentation 1972; IIR §11.4.3.

RETRIEVAL · BM25, WORKED

BM25 on last week's example: D3 now beats D2.

DOC	TERMS AFTER STOP LIST + STEMMING	|D|	TF INFO	TF RETRIEV	BM25
D1	info · retriev · VS · retriev · method	5	1	2	1.09
D2	info · theory · form · basis · probabil · method	6	1	0	0.42
D3	retriev · money · safe	3	0	1	0.55

THE ARITHMETIC

N = 3, avgdl = 4.67, k_1 = 1.2, b = 0.75. Both query terms appear in 2 of 3 documents, so idf = ln(1 + 1.5/2.5) = 0.47 for each.

D1's second “retriev” adds only 0.38 to the term's weight (1.35 vs 0.97). D3 is short (3 terms vs 4.67 on average), so its single match is boosted.

WHAT CHANGED

D1 1.09

D3 0.55

D2 0.42

BM25 fixed counting and length. It made D3 — a story about money — beat D2. Words still aren't meaning. That is Section 3.

SOURCE Calculation by the lecturer with the BM25 formula on the previous slide (Lucene-style idf); documents from ITMO SMA Module 2.

RETRIEVAL · LEARNING TO RANK

Web search blends hundreds of signals with a learned model.

Text match

BM25 on title, body, anchor text and URL.

Links

PageRank and anchor text from other pages (Part 2).

Freshness

News and events need recent pages; reference pages don't.

Place & language

“Pizza” means a different list in Moscow and in Singapore.

Behaviour

Clicks and dwell time — biased: top results get clicked anyway.

Quality & spam

Site reputation, link farms, AI-generated filler.

THREE WAYS TO LEARN A RANKING

- ✓ **Pointwise** predict each document's relevance grade.
- ✓ **Pairwise** learn which of two documents is better (RankNet, 2005).
- ✓ **Listwise** optimise the whole list's metric. LambdaMART ensembles won the 2010 Yahoo! Learning to Rank Challenge.

Section 4 reuses all three — a feed is a ranking too.

SOURCE Google, Guide to Google Search ranking systems (Dec 2025); Burges et al., ICML 2005 (RankNet); Burges, MSR-TR-2010-82 (LambdaMART); Joachims et al., SIGIR 2005 (click bias).

RETRIEVAL · DID IT WORK?

Order matters: precision@k, MRR and nDCG.

ONE QUERY, FIVE RESULTS, GRADED 0–3

#1	not relevant	○ ○ ○
#2	relevant	● ● ●
#3	relevant	● ● ○
#4	not relevant	○ ○ ○
#5	relevant	● ○ ○

0.60

PRECISION@5

3 of the top 5 are relevant. Ignores order within the five.

0.50

MRR

1 / rank of the first relevant result. The first hit is at #2.

0.67

nDCG@5

Graded gains, discounted by $\log_2$ (rank + 1), divided by the ideal order's score.

nDCG, STEP BY STEP

DCG = $7/\log_2 3 + 3/\log_2 4 + 1/\log_2 6 = 6.30$, with gain $2^{\text{grade}} - 1$. Ideal order (3, 2, 1, 0, 0) scores 9.39. nDCG = 6.30 / 9.39 = 0.67. Swap #1 and #2 — same five results — and nDCG jumps to 0.95.

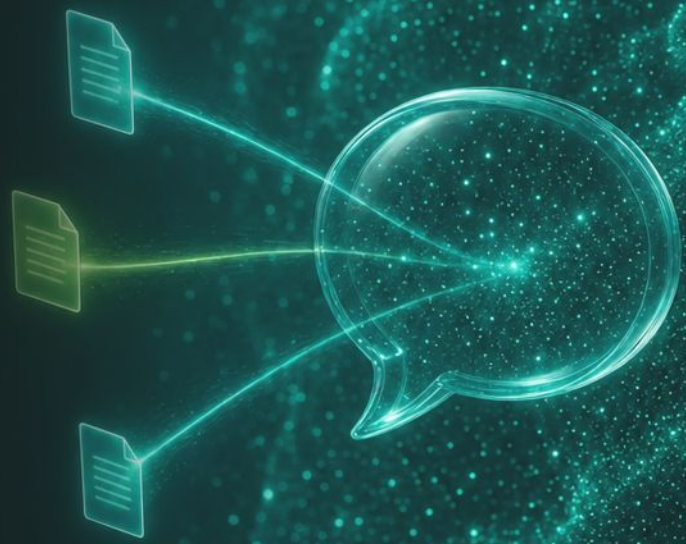
SOURCE Järvelin & Kekäläinen, ACM TOIS 2002 (nDCG); IIR ch. 8. Calculation by the lecturer; the full metrics toolkit comes in the evaluation lecture.

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03

SECTION 3

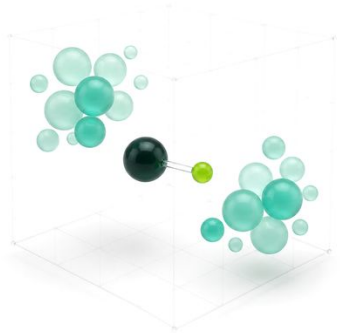
Neural search and RAG



PEW RESEARCH CENTER · JULY 2025

“With an AI summary on the page, users clicked a result on 8% of visits — versus 15% without one.”

Search is turning from a list of links into an answer. Underneath, it is still retrieval: embeddings find the evidence, an LLM writes it up.



NEURAL SEARCH · THE VOCABULARY GAP

Words don't match. Meanings do.

LEXICAL MISS

Same need, no shared word

“cheap flights to Singapore” vs “low-cost airfare to Changi”: zero terms in common, so BM25 scores 0.

LEXICAL FALSE HIT

Shared word, different need

“information retrieval” vs “he retrieved his money”: the stem matches, the meaning doesn't.

EMBEDDING SPACE (2-D SKETCH)



Illustrative positions, not real coordinates.

THE IDEA

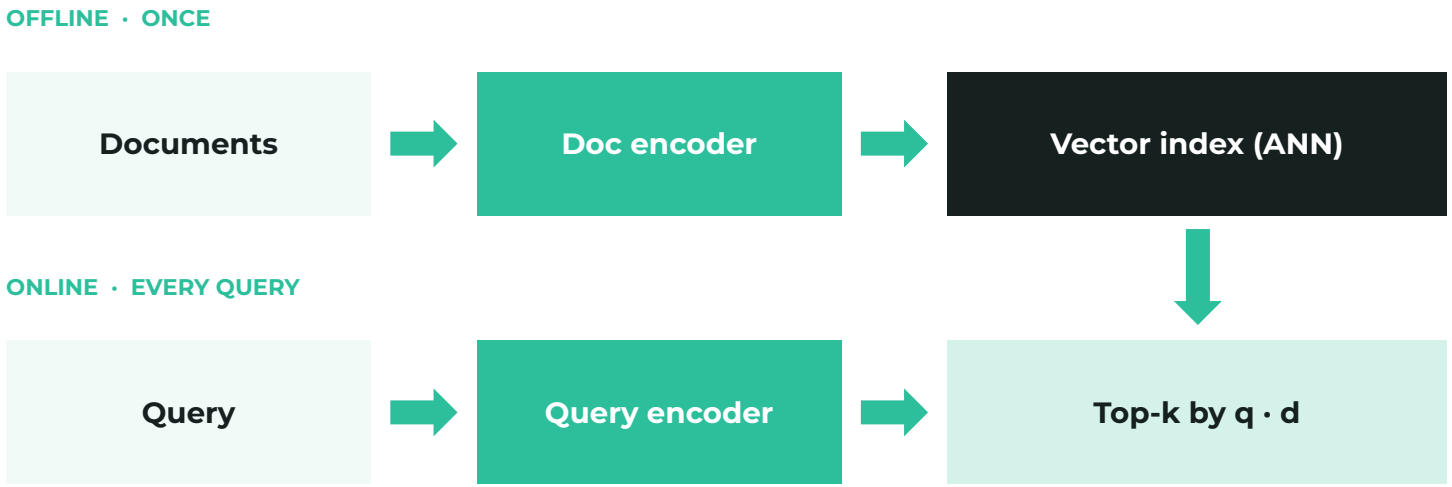
Embed queries and documents in the same space.

Similar meaning lands nearby (Lecture 2: “you shall know a word by the company it keeps”). Search becomes: find the nearest document vectors to the query vector.

SOURCE Farseev, ITMO SMA Lecture 2 (embeddings); Firth 1957. Positions on the sketch are illustrative.

NEURAL SEARCH · DENSE RETRIEVAL

Dense retrieval: embed once, search by nearest neighbour.



- **Cheap per query:** one encoder pass plus a vector look-up; documents were embedded in advance.
- **Weak spot:** rare names, product codes and numbers can blur into the vector.

DPR · EMNLP 2020

78.4%

vs 59.1% for BM25

Top-20 retrieval accuracy on Natural Questions: is a passage with the answer among the first 20? Two BERT encoders, trained on question–passage pairs, beat a tuned BM25 by 19 points.

SOURCE Karpukhin et al., “Dense Passage Retrieval for Open-Domain Question Answering”, EMNLP 2020 (Facebook AI, UW, Princeton).

NEURAL SEARCH · HYBRID SEARCH

BM25 is still hard to beat. Use both.

BEIR · NEURIPS 2021

18 datasets, zero-shot

Dense models trained on one dataset often lose to BM25 on others: legal, biomedical, tweets, arguments. The authors' summary: “BM25 is a robust baseline.”

Lexical match is precise on names and codes; dense match finds paraphrases. They fail differently — so combine them.

RECIPROCAL RANK FUSION

$$RRF(d) = \Sigma\ 1 / (60 + rank(d))$$

DOC	BM25	DENSE	RRF
B	2	1	0.0325
A	1	3	0.0323
D	4	2	0.0318
C	3	–	0.0159
E	–	4	0.0156

WHY RRF

- **No calibration.** Uses ranks only, so BM25 scores and cosine scores never need to be compared.
- **Agreement wins.** B is top-2 in both lists and beats A, which only one list loved.
- **k = 60** damps the top ranks; the value from the original paper.

SOURCE Thakur et al., BEIR, NeurIPS 2021 Datasets & Benchmarks; Cormack, Clarke & Büttcher, SIGIR 2009 (RRF). Fusion example calculated by the lecturer.

NEURAL SEARCH · RE-RANKING

Fast to retrieve, slow to judge: three encoders.

BI-ENCODER

Read separately

Query and document are embedded independently; score = dot product. Documents pre-computed. Fast over millions, but coarse.

LATE INTERACTION · COLBERT

Match token by token

Keep one vector per token; each query token finds its best document token (MaxSim). About 100× faster than BERT rerankers (Stanford, SIGIR 2020).

CROSS-ENCODER

Read together

Query and document go through the transformer as one input, attending to each other. Most accurate — and too slow for more than ~100 candidates.

A TYPICAL 2026 RETRIEVAL STACK

Corpus
1B passages

BM25 + dense
1,000

Cross-encoder
top 50

LLM reads
top 5

SOURCE Khattab & Zaharia, “ColBERT”, SIGIR 2020 (two orders of magnitude faster than BERT re-rankers); Nogueira & Cho 2019 (BERT re-ranking). Stack sizes are typical, not fixed.

NEURAL SEARCH · NEAREST NEIGHBOURS AT SCALE

The nearest of a billion vectors: walk a small-world graph.



Greedy search: start at the top, hop toward the query, drop a layer, repeat.

WHY IT MATTERS

- ✓ **Exact search** compares the query with every vector: O(N) per query.
- ✓ **HNSW** a layered small-world graph: about log N hops to the neighbourhood.
- ✓ **FAISS (Meta, 2017)** billion-scale similarity search on GPUs.
- ✓ **The trade** a little recall for orders-of-magnitude speed. Tune it, measure it.

Small worlds return in Part 2, next week.

SOURCE Malkov & Yashunin, HNSW (arXiv 2016; IEEE TPAMI 2020); Johnson, Douze & Jégou, “Billion-scale similarity search with GPUs” (2017).

NEURAL SEARCH · RETRIEVAL-AUGMENTED GENERATION

RAG: let the LLM read before it writes.



01

Chunk & embed

Split documents into passages; embed and index them. Offline.

02

Retrieve

Embed the question; fetch the top-k passages (hybrid + rerank).

03

Augment

Paste the passages into the prompt, with their sources.

04

Generate

The LLM answers from the passages and cites them.

WHY TEAMS USE IT

- **Fresh.** Update the index, not the model.
- **Private.** Your documents never enter training.
- **Checkable.** Every claim points to a passage.

Lewis et al. (Facebook AI, NeurIPS 2020) named it: parametric memory (the model's weights) plus non-parametric memory (a dense index of Wikipedia, read by a neural retriever).

SOURCE Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”, NeurIPS 2020; Stanford CS224N (Winter 2026), Lecture 10: Agents, Tool Use, and RAG.

NEURAL SEARCH · CASE: LEGAL RAG, STANFORD

Retrieval reduces hallucination. It doesn't remove it.

17%

LEXIS+ AI

of answers hallucinated — more than 1 in 6. Ask Practical Law AI: similar.

33%

WESTLAW AI RESEARCH

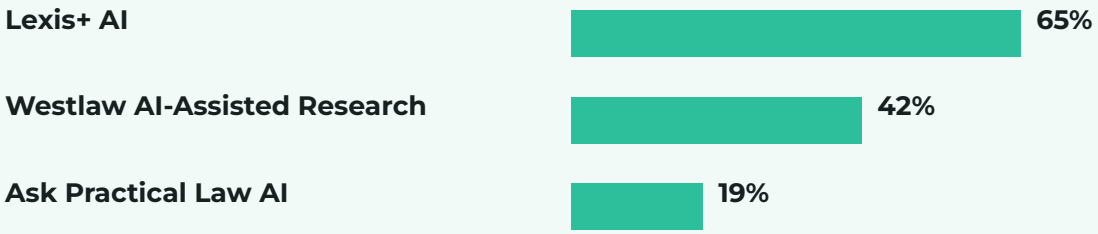
of answers hallucinated — 1 in 3.

202

LEGAL QUERIES

tested by Stanford's RegLab and HAI, scored by hand.

ACCURATE AND COMPLETE ANSWERS



LESSONS FOR ANY RAG SYSTEM

- ✓ **Retrieval bounds the answer.** Wrong passage in, confident wrong answer out — with a real-looking citation.
- ✓ **Test the parts.** Was the right passage retrieved? Did the answer actually use it?
- ✓ **Marketing ≠ measurement.** Vendors had called these tools “hallucination-free”.

SOURCE Magesh, Surani, Dahl, Suzgun, Manning & Ho, “Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools”, J. Empirical Legal Studies 22 (2025).

NEURAL SEARCH · 2026 PATTERNS

Four ways RAG grew up.

01

Hybrid + rerank

BM25 and dense retrieval, fused, then a cross-encoder. The default production stack.

02

GraphRAG

An LLM builds an entity graph and community summaries, so it can answer “what are the themes?” about a whole corpus (Microsoft, 2024).

03

Agentic search

The LLM writes its own queries, reads, and searches again until it is satisfied. Lecture 8.

04

Generative retrieval

The model memorises the corpus and generates document IDs directly (DSI, Google, NeurIPS 2022). Returns in Section 4.

Does a long context window kill retrieval? No. A million-token window cannot hold hundreds of billions of pages, and every token read costs time and money. Retrieval decides what is worth reading.

SOURCE Edge et al., “From Local to Global: A Graph RAG Approach” (Microsoft Research, 2024); Tay et al., “Transformer Memory as a Differentiable Search Index”, NeurIPS 2022.



NEURAL SEARCH · WHAT AI ANSWERS DO TO THE WEB

AI answers take the click.

8% vs 15%

CLICKS ON A RESULT

With vs without an AI summary on Google (Pew, March 2025 browsing data, 900 US adults).

1%

CLICKS INSIDE THE SUMMARY

Links in the AI summary itself were almost never clicked.

−33%

ORGANIC GOOGLE TRAFFIC

To 2,576 publisher sites worldwide, Nov 2024 → Nov 2025 (Chartbeat).

−40%

GOOGLE REFERRALS

To publishers, July 2025 → July 2026 — after −22% the year before (Chartbeat).

FOR MARKETERS

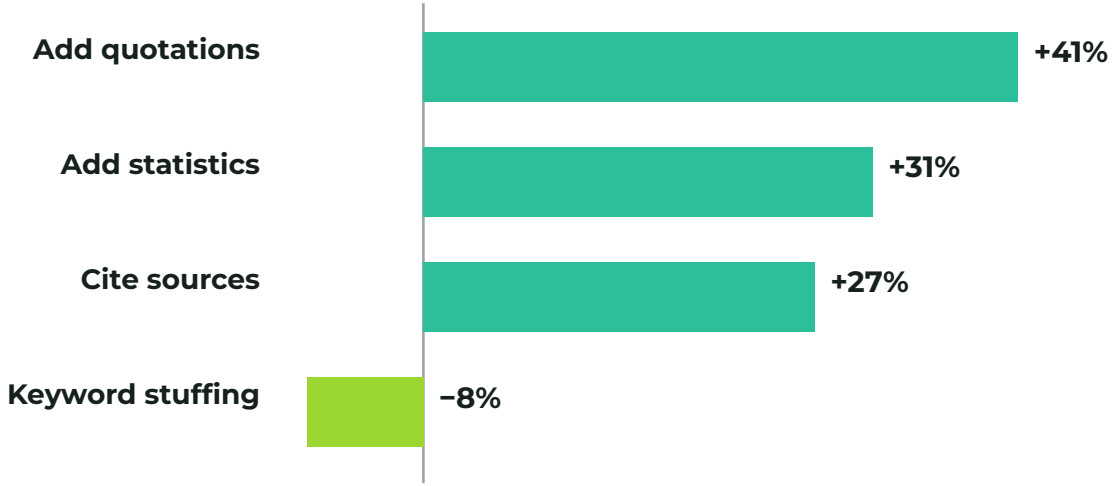
Ranking first is no longer the finish line. The new question is whether a model retrieves your page, trusts it, and quotes it in the answer — because many users never scroll to the links.

SOURCE Pew Research Center (22 Jul 2025); Reuters Institute, Trends and Predictions 2026 (Chartbeat data); Chartbeat 2026 Publisher Playbook via Search Engine Journal (24 Sep 2026).

NEURAL SEARCH · GENERATIVE ENGINE OPTIMISATION

GEO: write to be retrieved and quoted.

CHANGE IN VISIBILITY INSIDE AI ANSWERS (GEO-BENCH, 10,000 QUERIES)



Relative change in position-adjusted word count of the source in the generated answer. The best methods lifted visibility by up to 40%; tested live on Perplexity, up to 37%.

WHAT IT MEANS

- ✓ **Evidence gets quoted.** Answer engines prefer checkable facts, numbers and named sources.
- ✓ **Old SEO tricks backfire.** Keyword stuffing scored below doing nothing.
- ✓ **Sameness is a risk.** If every brand optimises for the same model, content converges — the homogenisation problem in this team's SIGIR 2026 paper.

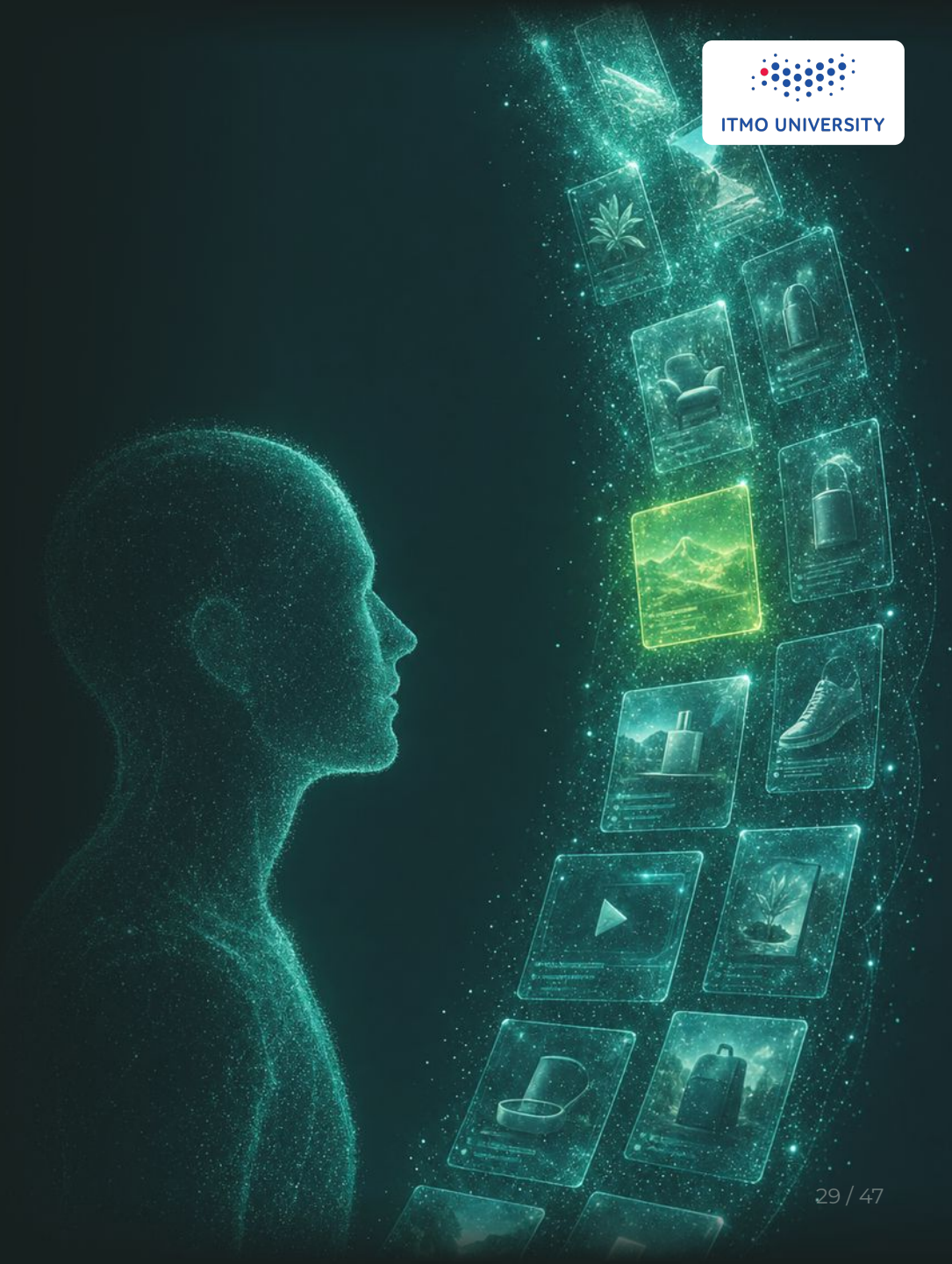
SOURCE Aggarwal et al., “GEO: Generative Engine Optimization”, KDD 2024; SOMIN, ACM SIGIR 2026 (countering content homogenisation).

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SECTION 4

Recommender systems



META · ICML 2024

“Meta's generative recommender has 1.5 trillion parameters — and its quality scales with compute like an LLM's.”

Recommendation began as filling blanks in a ratings table. In 2026 it is a sequence model that predicts your next action.



RECOMMENDATION · THE PROBLEM

Fill in the blanks of a mostly empty table.

THE UTILITY MATRIX · USERS × ITEMS

	Avatar	LOTR	Dune	Titanic	Notebook
Alice	5	4	4	1	2
Bob	4	5	3	2	1
Carol	?	5	4	1	2
David	2	1	2	5	4
Eve	1	2	1	4	5

A toy table is full. Real ones are not — and the job is to predict the cells nobody filled.

NETFLIX PRIZE DATA, 2006

1.2%

of cells filled: 100.5M ratings in a 480,189 × 17,770 table (8.5 billion cells).

- **Explicit vs implicit.** Stars are rare; views and clicks are many — but carry no clear negatives.
- **Cold start.** A new user is an empty row; a new item, an empty column.
- **Long tail.** Most items have almost no data; a few have most of it.

SOURCE Leskovec, Stanford CS246 “Recommender Systems I” (Winter 2026); Netflix Prize dataset (Bennett & Lanning, KDD Cup 2007): 100,480,507 ratings.

RECOMMENDATION · APPROACHES

Three classic ways to fill the table.

01 · POPULARITY

What most people liked

- ✓ Trending, top-10, best-sellers.
- ✓ A strong baseline for new users.
- ✓ No personalisation; feeds the rich-get-richer loop.

02 · CONTENT-BASED

Items like the ones you liked

- ✓ Uses item features: tf.idf, embeddings, tags.
- ✓ Works for brand-new items.
- ✓ Over-specialises: more of the same.

03 · COLLABORATIVE FILTERING

People like you liked it

- ✓ Needs only the interaction table.
- ✓ Finds non-obvious matches.
- ✓ Cold start for new users and items; popularity bias.

Hybrid in practice: every production system blends all three — popularity for cold start, content for new items, collaborative signals for everything else.

SOURCE Leskovec, Rajaraman & Ullman, Mining of Massive Datasets, ch. 9; Stanford CS246 (Winter 2026); ITMO SMA Module 9 (venue recommendation strategies).

RECOMMENDATION · CONTENT-BASED

Content-based: your profile is the average of what you liked.

01

Describe items

Each post becomes a vector: tf.idf from Section 2, or an embedding.

02

Build the profile

Average the vectors of the posts you liked (weight recent ones more).

03

Score candidates

Cosine between your profile and each new post.

04

Rank and show

Top-k — then diversify, so it isn't ten copies of one post.

EXAMPLE · A LINKEDIN READER

You liked three posts on AI in marketing. Your profile vector leans toward “LLM”, “campaign”, “brand”. A new post on generative ads scores 0.8; one on Kubernetes scores 0.1. The new post can be recommended the minute it is published — no one has to like it first.

TRADE-OFFS

- ✓ **Good:** new items, transparent reasons.
- ✓ **Bad:** the filter bubble by design; needs good item features.

SOURCE Leskovec, Rajaraman & Ullman, MMDS §9.2; Stanford CS246 RecSys I. Example scores are illustrative.

RECOMMENDATION · COLLABORATIVE FILTERING

Item-item CF: rate a film by its most similar films.

STEP 1 · SIMILARITY OF AVATAR TO CAROL'S FILMS

FILM	CAROL	RAW COSINE	CENTRED COSINE
Dune	4	0.996	+0.87
LOTR	5	0.957	+0.75
Titanic	1	0.587	−0.92
Notebook	2	0.587	−0.92

Centre each user's ratings on their own mean first. Raw cosine calls Titanic “0.59 similar” to Avatar only because all ratings are positive numbers.

STEP 2 · WEIGHTED AVERAGE OF THE TOP-2 NEIGHBOURS

$$\hat{r} = (0.87 \times 4 + 0.75 \times 5) / (0.87 + 0.75)$$

4.5

Carol's predicted rating for Avatar (4.46).

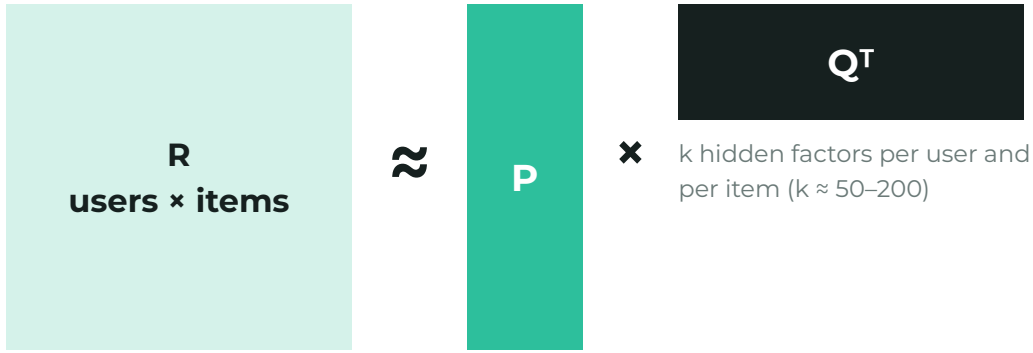
WHY ITEM-ITEM

Items have more ratings than users and their similarities change slowly, so they can be pre-computed. Amazon's “customers who bought this also bought” made item-to-item CF famous (2003).

SOURCE Sarwar et al., WWW 2001 (item-based CF); Linden, Smith & York, IEEE Internet Computing 2003 (Amazon); calculation by the lecturer on the table from the previous slides.

RECOMMENDATION · MATRIX FACTORISATION

Matrix factorisation: hidden tastes, learned from the gaps.



0.9514

NETFLIX'S RMSE, 2006

Cinematch on the quiz set. The prize: $1M for 10% better.

0.8567

WINNER, SEPT 2009

BellKor's Pragmatic Chaos: 10.06% better, submitted 20 minutes before a tied team.

LEARN P AND Q FROM KNOWN RATINGS ONLY

$$\min \Sigma (r_{ui} - p_u \cdot q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2)$$

Gradient descent or alternating least squares. λ stops the model memorising the few ratings it has. Prediction: $\hat{r} = p_u \cdot q_i$.

TWO LESSONS FROM THE PRIZE

- ✓ **Engineering beats decimals.** Netflix shipped SVD and RBMs, not the 100-model blend: the extra accuracy wasn't worth the cost.
- ✓ **"Anonymised" isn't.** Researchers re-identified users; the sequel prize was cancelled in 2010.

SOURCE Koren, Bell & Volinsky, IEEE Computer 2009; Netflix Prize (2006–09); Amatriain & Basilico, Netflix Tech Blog (Apr 2012); Narayanan & Shmatikov, IEEE S&P 2008.

RECOMMENDATION · WHAT TO OPTIMISE

Pick the target and you pick the behaviour.

2006

Ratings

Predict stars (Netflix Prize, RMSE). People rarely rate, and stars aren't what they watch.

2010s

Clicks

Cheap and plentiful. Rewards clickbait thumbnails and headlines.

2016

Watch time

YouTube ranked by expected watch time per impression: a click you leave after 3 seconds counts little.

2026

Many actions

X's open-source ranker predicts likes, replies, reposts, dwell — and blocks, mutes, reports. The score is a weighted sum.

Goodhart's law: when a measure becomes a target, it ceases to be a good measure. Every objective is a bet about what users value — and the system will exploit its gaps.

SOURCE Covington, Adams & Sargin, RecSys 2016; xAI, github.com/xai-org/x-algorithm (Jan 2026); Netflix Prize; Goodhart (1975), as phrased by Strathern (1997).



RECOMMENDATION · INDUSTRIAL ARCHITECTURE

Two towers retrieve. A big model ranks.

USER TOWER

history · profile · context

ITEM TOWER

id · text · image · creator



user vector · item vector → dot product

Item vectors are pre-computed and indexed; at request time only the user tower runs, then an ANN look-up.

THEN RANK

A much bigger model scores each of the few thousand candidates with rich features and cross-features, predicting several outcomes at once. A re-ranker adds diversity and rules.

IN PRODUCTION

- ✓ **YouTube, 2016:** "one network for candidate generation and one for ranking" — millions of videos to hundreds.
- ✓ **Meta Andromeda, 2024:** ad retrieval with 10,000× more model capacity; +6% recall, +8% ads quality on selected segments.
- ✓ **X Phoenix, 2026:** a two-tower model finds out-of-network posts for the transformer ranker.

SOURCE Covington, Adams & Sargin, RecSys 2016; Meta Engineering, "Meta Andromeda" (2 Dec 2024); xAI, x-algorithm repository (20 Jan 2026).

RECOMMENDATION · CASE: X'S OPEN-SOURCE FEED

Inside X's “For You” feed: the code is public.

SOURCES

Thunder: posts from accounts you follow. Phoenix retrieval: out-of-network posts via two towers.

RANK

Phoenix: a transformer with Grok's architecture predicts many actions per post.

SCORE

Weighted sum of predicted actions; blocks, mutes and reports count against.

FILTER

Rules, policy and a diversity re-ranker (added Aug 2026).

THEN AND NOW

	MARCH 2023	JANUARY 2026
Candidates	~1,500 from ~500M daily posts	Thunder + Phoenix retrieval
Ranker	~48M-parameter neural net	Grok-architecture transformer
Features	Hand-engineered + embeddings	Hand-engineered features removed

WHY STUDY IT

Every stage of today's funnel in one repository you can read: github.com/xai-org/x-algorithm. Code shows the design — not the weights or the data.

SOURCE Twitter Engineering blog (31 Mar 2023); xAI, github.com/xai-org/x-algorithm (Apache-2.0, 20 Jan 2026; README update 13–14 Aug 2026); Decrypt (Jan 2026).

RECOMMENDATION · GENERATIVE RECOMMENDERS

From “what do you like?” to “what will you do next?”

2018

SASRec

Self-attention over your action history predicts the next item.

2024

Meta HSTU

1.5T parameters; +12.4% in online A/B tests; quality follows a power law in compute.

2025

Netflix & LinkedIn

Netflix: one foundation model on hundreds of billions of interactions. LinkedIn 360Brew: 150B-parameter decoder, 30+ tasks.

2025–26

Meta GEM (ads)

+5% Instagram and +3% Facebook ad conversions in Q2 2025; +18.9% conversions over the past year.

The shift: your history is a sentence and every action is a token. Recommendation becomes next-token prediction — and inherits the LLM's scaling laws.

SOURCE Kang & McAuley, ICDM 2018; Zhai et al., ICML 2024; Netflix Tech Blog (Mar 2025); LinkedIn, 360Brew, arXiv 2501.16450 (2025); Meta Engineering (10 Nov 2025; 13 Oct 2026).

RECOMMENDATION · SEMANTIC IDS

Semantic IDs: give every item words an LLM can generate.

HOW AN ITEM GETS A NAME

Item	Embedding	Quantise	Semantic ID
video, song, ad	content vector	RQ-VAE, 3 levels	<17> <203> <88>

Similar items share prefixes (<17> <203> …), so a transformer can generate the ID of the next item token by token — even for an item nobody has watched yet.

TIGER · GOOGLE, NEURIPS 2023	3 codewords from codebooks of 256; helps cold-start items.
PLUM · YOUTUBE, 2025	Gemini-based generative retrieval in production; Shorts CTR +4.96%.
GLIDE · SPOTIFY, 2026	Podcast discovery; new-show streams +14.3% in a 21-day A/B test.

SEARCH AND RECOMMENDATION MEET

Section 3: generate a document's ID.
Section 4: generate an item's ID.

One model, one vocabulary of items, two jobs. Meta described semantic-ID tokens for its ads models this week (19 Oct 2026). The funnel from Section 1 may collapse into a single generative model.

SOURCE Rajput et al., NeurIPS 2023 (TIGER); Google/YouTube, arXiv 2510.07784 (PLUM, 2025); Spotify, arXiv 2603.17540 (GLIDE, 2026); Meta Engineering, “GenRecs tokenization” (19 Oct 2026).

RECOMMENDATION · CASE FROM THIS COURSE

Where will she go next? Combine weak signals.



NEXT-PLACE PREDICTION · ICDM 2012

She checks in at a flower shop on a Saturday morning. Where next?

35M

Foursquare check-ins over 5 months in 2010. One right answer among thousands of venues, and very few check-ins per user.

SIGNALS TESTED

Popularity check-ins by anyone	**History** places she visited before	**Social** where her friends go
Category café after flowers?	**Distance** how far from here	**Time** hour and day of week

- ✓ **Personal signals win short lists:** history and friends beat popularity and distance for the top few guesses.
- ✓ **Combined beats single:** a supervised model over all features was best throughout.
- ✓ **Time matters:** noon is easier to predict than the evening. Today's delivery and map apps solve the same problem.

SOURCE Noulas, Scellato, Lathia & Mascolo, “Mining User Mobility Features for Next Place Prediction in Location-based Services”, IEEE ICDM 2012; Farseev, ITMO SMA Module 9.

RECOMMENDATION · COLD START

New users, new items: explore before you exploit.

01

New user

Ask a few onboarding questions; use context (device, place, time); start with what is popular nearby.

02

New item

Describe it by content: embeddings or semantic IDs place it next to items with history.

03

Bandits

Spend some impressions on uncertain items to learn. Yahoo!'s contextual bandit lifted news clicks 12.5%.

04

LLMs

Recommend zero-shot from a text description of the user. Good for cold start; weak on catalogue and freshness.

EXPLORE VS EXPLOIT

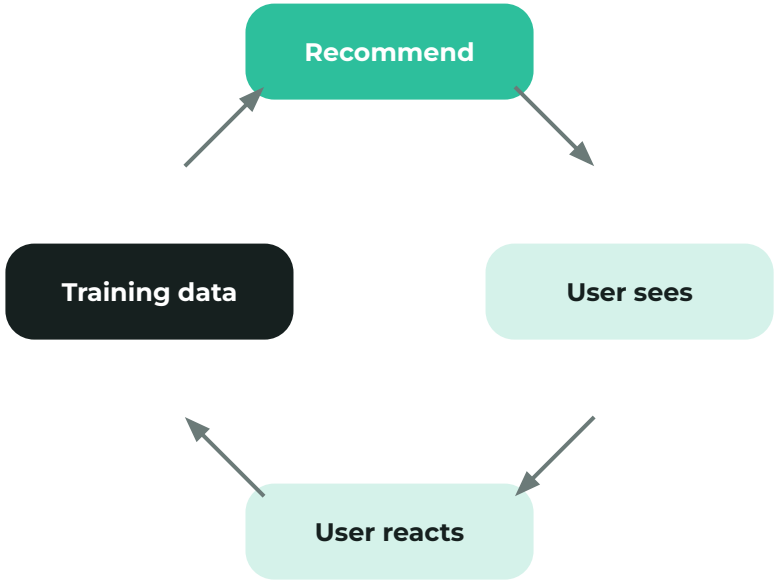
Showing only the current best item means you never learn whether something else is better. A little exploration costs a few clicks today and buys a better model tomorrow — and it is the honest answer to the rich-get-richer loop.

SOURCE Li, Chu, Langford & Schapire, “A Contextual-Bandit Approach to Personalized News Article Recommendation”, WWW 2010; Rajput et al., NeurIPS 2023.

RECOMMENDATION · FEEDBACK LOOPS

The recommender trains on its own output.

THE LOOP



SIMULATION · RECSYS 2018

Sameness without benefit

Feedback loops homogenise what users consume without making them better off.

TWITTER AUDIT · PNAS NEXUS 2025

More anger than chronological

Engagement ranking showed more anger (+0.47 SD) and out-group animosity (+0.24 SD); users liked its political picks less.

FACEBOOK & INSTAGRAM EXPERIMENT · SCIENCE 2023

Switch the algorithm off — attitudes barely move

A chronological feed for three months of the 2020 US election cut time on the platform sharply and showed more political and untrustworthy content, but polarisation did not change significantly. Lecture 1's Stanford study moved attitudes by reranking. Measure; don't assume.

SOURCE Chaney, Stewart & Engelhardt, RecSys 2018; Milli et al., PNAS Nexus 4(3), 2025; Guess et al., Science 381 (2023); Piccardi et al., Science (Nov 2025).

RECOMMENDATION · THE RULES IN 2026

Regulators now audit the ranking, not just the content.

EU DSA · ART. 27

Explain the ranking

Every platform must state its recommender's main parameters in plain language, and let users switch between options.

EU DSA · ART. 38

Offer a non-profiled feed

Very large platforms must provide at least one recommender option that is not based on profiling.

EU · FEB & JUL 2026

Addictive design

Preliminary findings against TikTok, then Instagram and Facebook, cite the personalised recommender.

US · JAN 2026

TikTok's algorithm moves

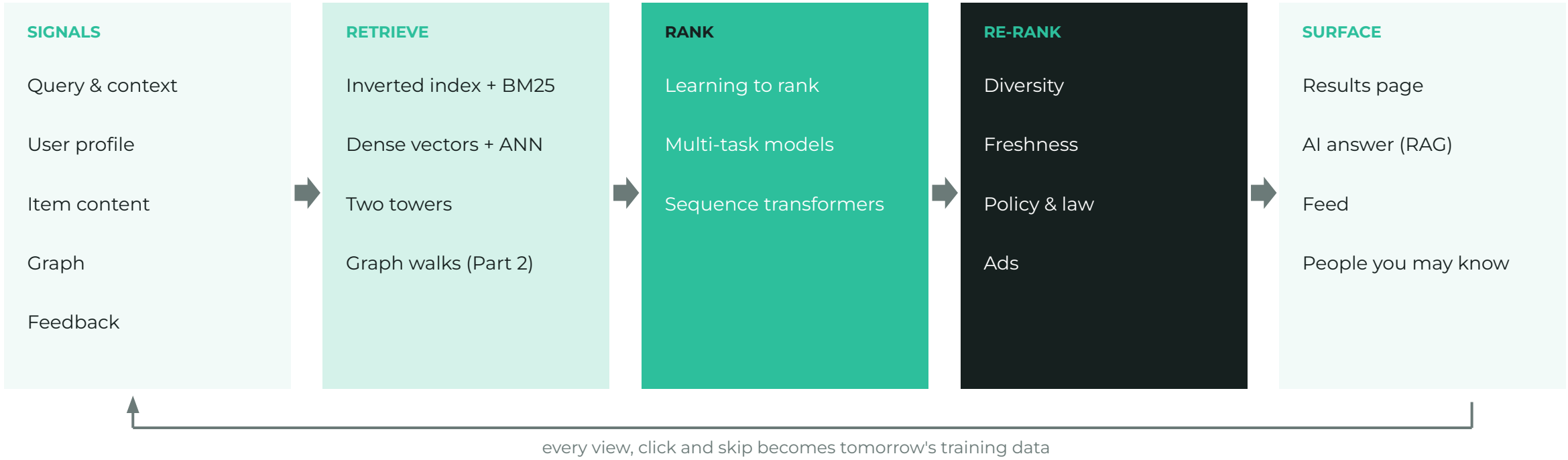
The new US joint venture retrains the recommender solely on US data, in Oracle's US cloud.

For builders: log why each item was shown, keep a non-personalised fallback, and be able to explain your top three ranking signals to a regulator — or a journalist.

SOURCE Regulation (EU) 2022/2065 (DSA) Arts. 27, 38; European Commission press releases (6 Feb 2026; 10 Jul 2026); TikTok USDS Joint Venture announcement (23 Jan 2026).

LECTURE 3, PART 1 · PUTTING IT TOGETHER

One stack: retrieve, rank, re-rank.



Takeaway: search, feeds and “people you may know” are one matching machine with different inputs.

COURSE · FROM LECTURE TO LAB

Use today's tools in Lab 1 and in your project.

LAB 1 · DUE SUN 25 OCT, 23:59

Did this post beat its author's usual?

- **Retrieve the author's history.** Find their most similar past posts with BM25 or embeddings; use how those did as features.
- **Measure typicality.** Distance between a post and the author's own profile vector.
- **Network features, carefully.** Reshares and mentions say more than followers.

MINI-PROJECT · TOPIC DUE SUN 8 NOV

One page per team.

Problem, user, data, methods. Teams of 2–3. Approved by Sun 15 Nov; Lab 2 due Sun 6 Dec.

PROJECT IDEAS FROM TODAY

- A RAG assistant for this course that cites the slide it used.
- “Who should see this post?” — a recommender for LinkedIn writers.
- Hybrid search over the Lab 1 posts — BM25 + embeddings, scored with nDCG.
- A GEO auditor: will an AI engine quote this page?

SOURCE Lab 1 brief (Lecture 1 appendix, slides 33–37); course schedule (Lecture 1, slide 30). Deadlines are 23:59 Moscow time.

COURSE · GO DEEPER

Free reading from Stanford, MIT and the papers behind today.

01 SEARCH & RAG

Stanford CS224N (Winter 2026) — Lecture 10: Agents, Tool Use, and RAG (BM25, DPR, ColBERT)

web.stanford.edu/class/cs224n

02 IR TEXTBOOK

Manning, Raghavan & Schütze — Introduction to Information Retrieval: ch. 1, 6, 11 (index, tf.idf, BM25)

nlp.stanford.edu/IR-book

03 RECOMMENDER SYSTEMS

Stanford CS246 (Winter 2026), J. Leskovec — Recommender Systems I & II

web.stanford.edu/class/cs246

04 RECOMMENDERS IN PRODUCTION

Covington, Adams & Sargin — Deep Neural Networks for YouTube Recommendations (RecSys 2016)

research.google/pubs/pub45530

05 GENERATIVE RECOMMENDERS

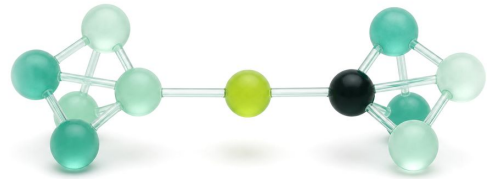
Zhai et al. — Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers (ICML 2024)

arxiv.org/abs/2402.17152

06 RAG IN PRACTICE

MIT 15.773 Hands-on Deep Learning (OCW, 2024) — Lecture 9: LLMs and Retrieval-Augmented Generation

ocw.mit.edu/courses/15-773-hands-on-deep-learning-spring-2024



OTHER DETAILS · NEXT LESSON

Next week: Lecture 3, Part 2 — networks and graphs.

PART 2 · WEDNESDAY 28 OCTOBER

The graph underneath search and feeds: small worlds, weak ties, homophily, influence and how things spread, PageRank and communities, fake networks, and graph neural networks.

- **Lab 1:** due this Sunday, 25 Oct, 23:59. Submit predictions and the notebook.
- **Mini-project topic:** one page per team by Sun 8 Nov, 23:59.
- **Office hours:** Prof. Aleks Farseev, 12 PM on Zoom, via Daniel Onishchenko or Jack Zhao.
- **Slides & reading:** posted on the course Telegram channel after class.

BEFORE PART 2

Audit your “People you may know”.

Open LinkedIn (or any app that suggests people). For the first ten suggestions, write down how many mutual connections each has, and whether you would call them close, an acquaintance, or a stranger. Bring the numbers.

SOURCE Course schedule (Lecture 1, slide 30). Part 2 replaces next week's slot; later lectures shift by one week [Confirm].